Module 8: Trees

Part 1: Basic Decision Trees

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Some lecture slides and instructional materials in this course are adapted from the following sources:

•*An Introduction to Statistical Learning: With Applications in R (Second Edition)*
Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani
Springer, 2021

•Online course materials developed by Trevor Hastie, Robert Tibshirani, and collaborators.

Tree-based methods

- Tree-based methods are supervised learning methods that can be used for regression and classification.
- These methods involve stratifying or segmenting the predictor space into a number of simple regions.
- Tree-based methods are simple and useful for interpretation.
- However, they typically are not competitive with the best supervised learning approaches.
- Instead, we use methods that involve combining multiple trees: bagging, random forests, and boosting.

Tree

- Recall we assign an observation in a given region to the most commonly occurring class of the training observations in that region.
- The classification error rate is just the fraction of training observations that do not belong to the most common class:

$$E = 1 - \max_k(\hat{p}_{mk}),$$

$\hat{p}_{mk}$ represents the proportion of training observations in the m th region that are from the k th class.

Turns out that classification error is not sufficiently sensitive for tree-growing, and in practice two other measures are preferred:

1. **Gini index**
2. **Entropy**

Tree

Response is qualitative (categorical): Classification Tree
Response is quantitative (numerical): Regression Tree

Gini index

Gini index is defined by

$$G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk}),$$

and can be interpreted as a measure of total variance across the K classes.

- Small values of Gini index are better.
- If all of the $\hat{p}_{mk}$ are close to 0 or 1 , then the Gini index takes on a small value.
- Therefore Gini index is also referred to as a measure of node purity.
- A small value of the Gini index indicates that a node contains predominantly observations from a single class.
- When k = 2, $G = 1 - P(Yes)^2$- $P(No)^2$

Entropy

Entropy is defined by

$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log(\hat{p}_{mk})$$

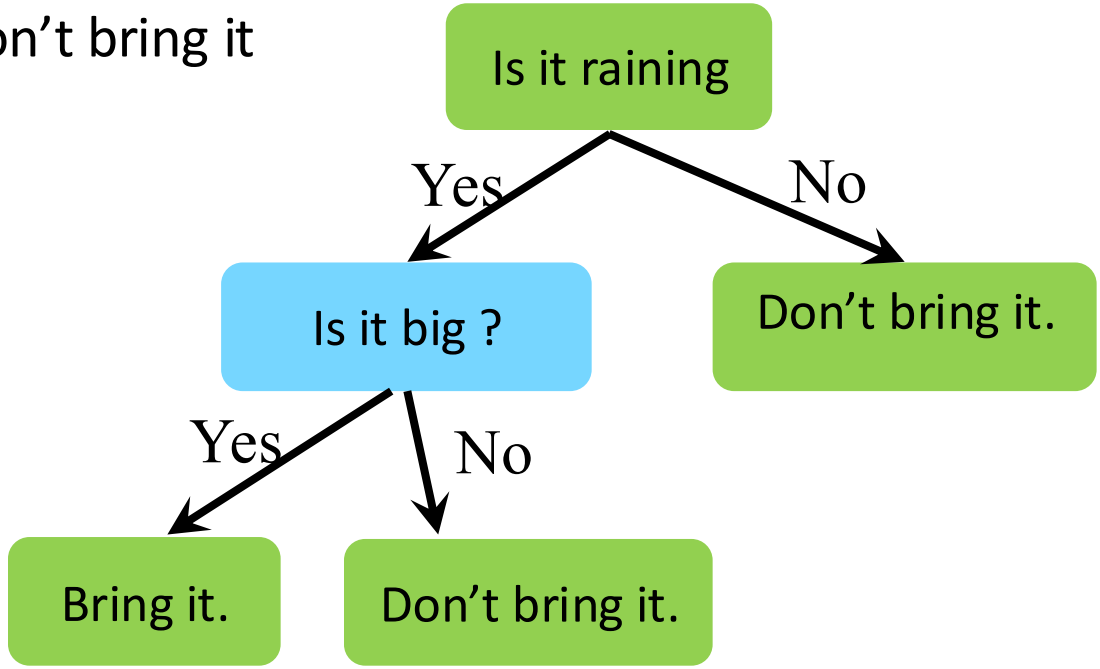
- It can be shown that D is always positive.
- Entropy will take on a small value if the $\hat{p}_{mk}$ are all near 0 or 1.
- Therefore, like the Gini index, the entropy will take on a small value if the m th node (or region) is pure.
- Gini index and entropy perform very similarly. Emerged in different disciplines.

How does classfication tree work ?

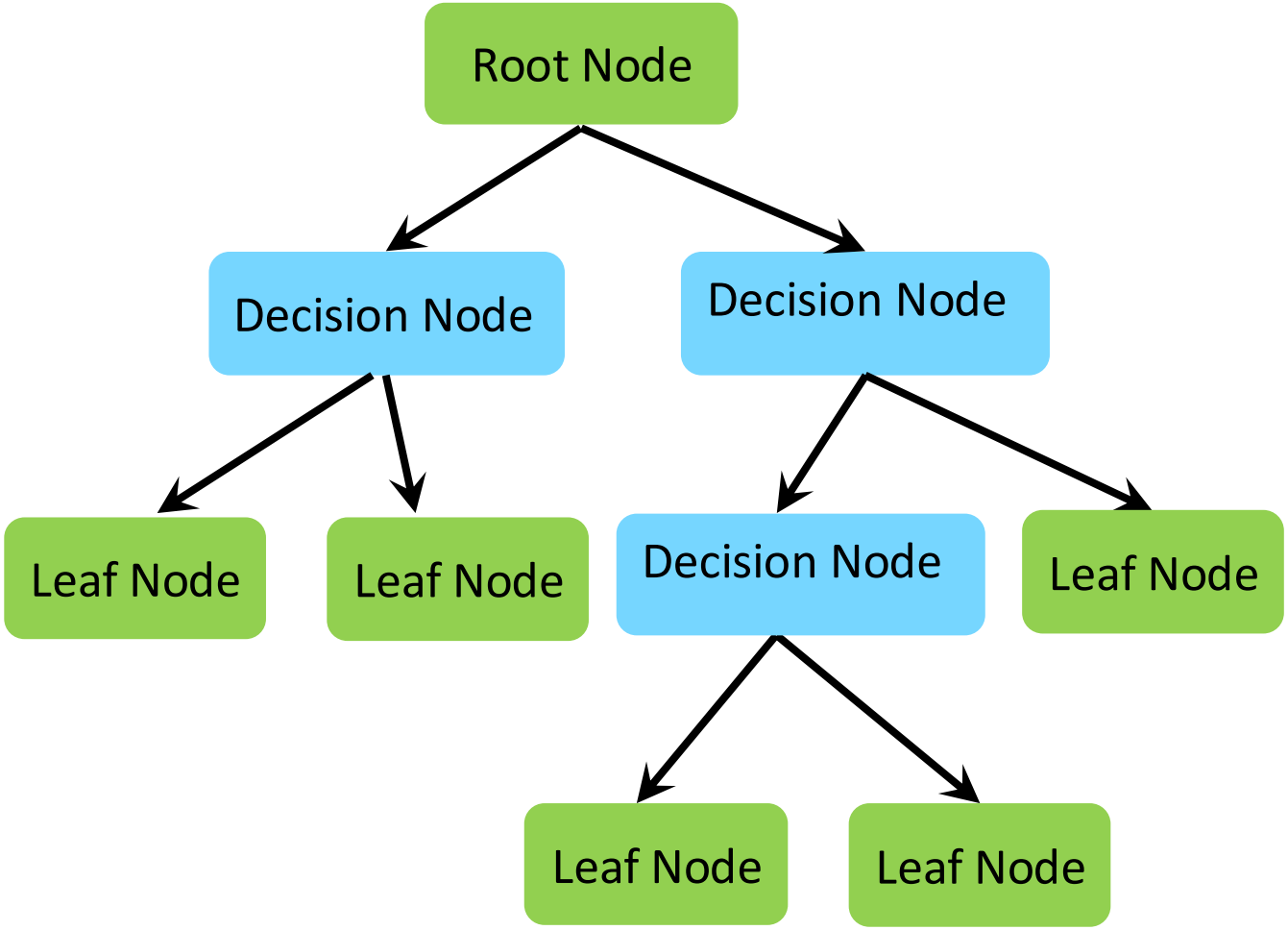
You're about to leave the house and need to decide:

Option 1: Bring an umbrella

Option 2: Don't bring it

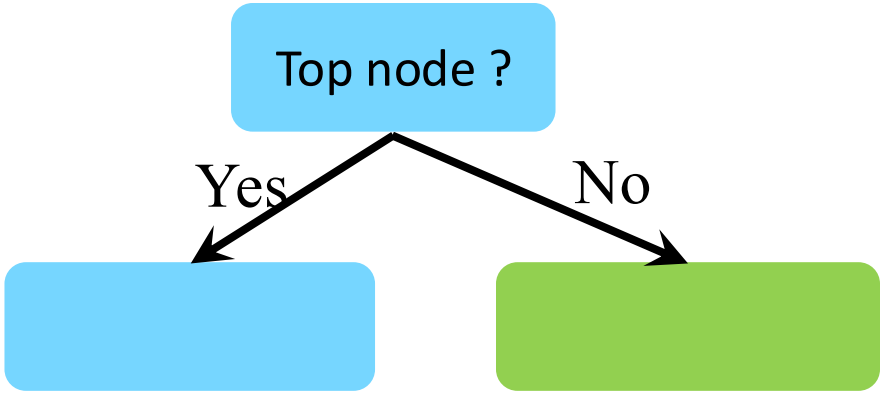


How does classfication tree work ?



Example: Predicting "Loves Cool Ice"

Loves Popcorn	Loves Soda	Age	Loves Cool As Ice
Yes	Yes	7	No
Yes	No	12	No
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



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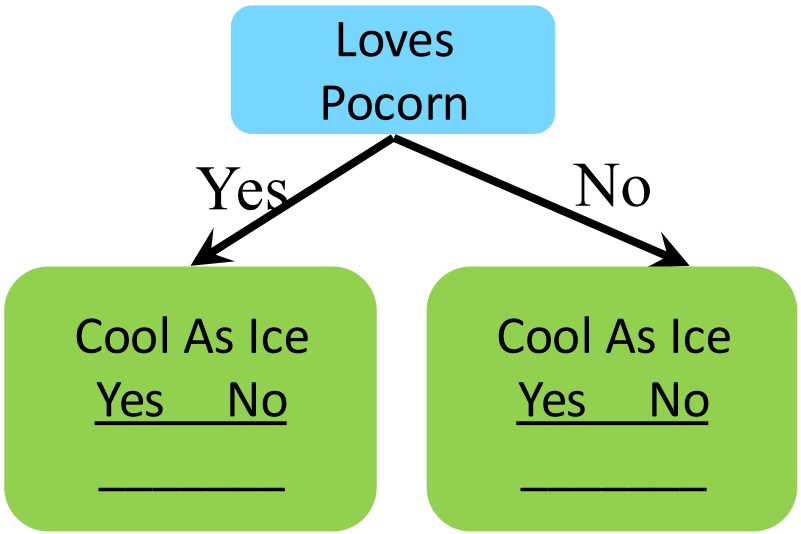
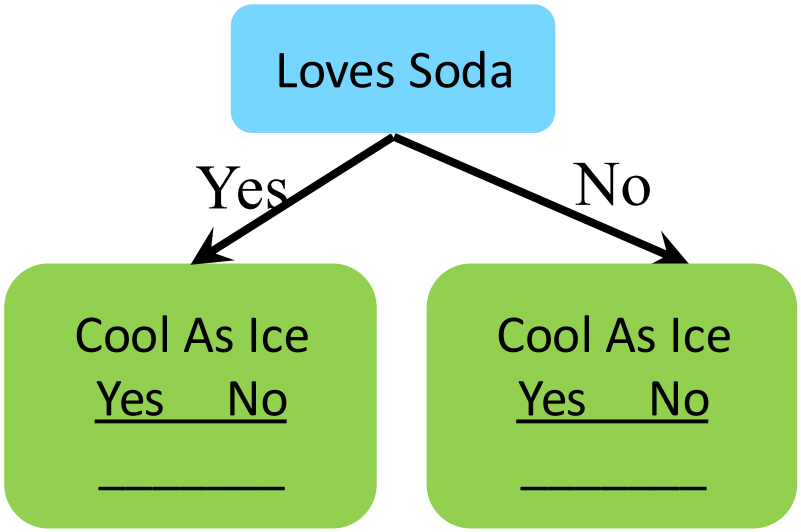




Example: Predicting "Loves Cool Ice"

Gini Impurity

= 1- (the probability of "Yes")^2
- (the probability of "No")^2

Gini impurity for leaf **Loves popcorn Yes**

= _____

=

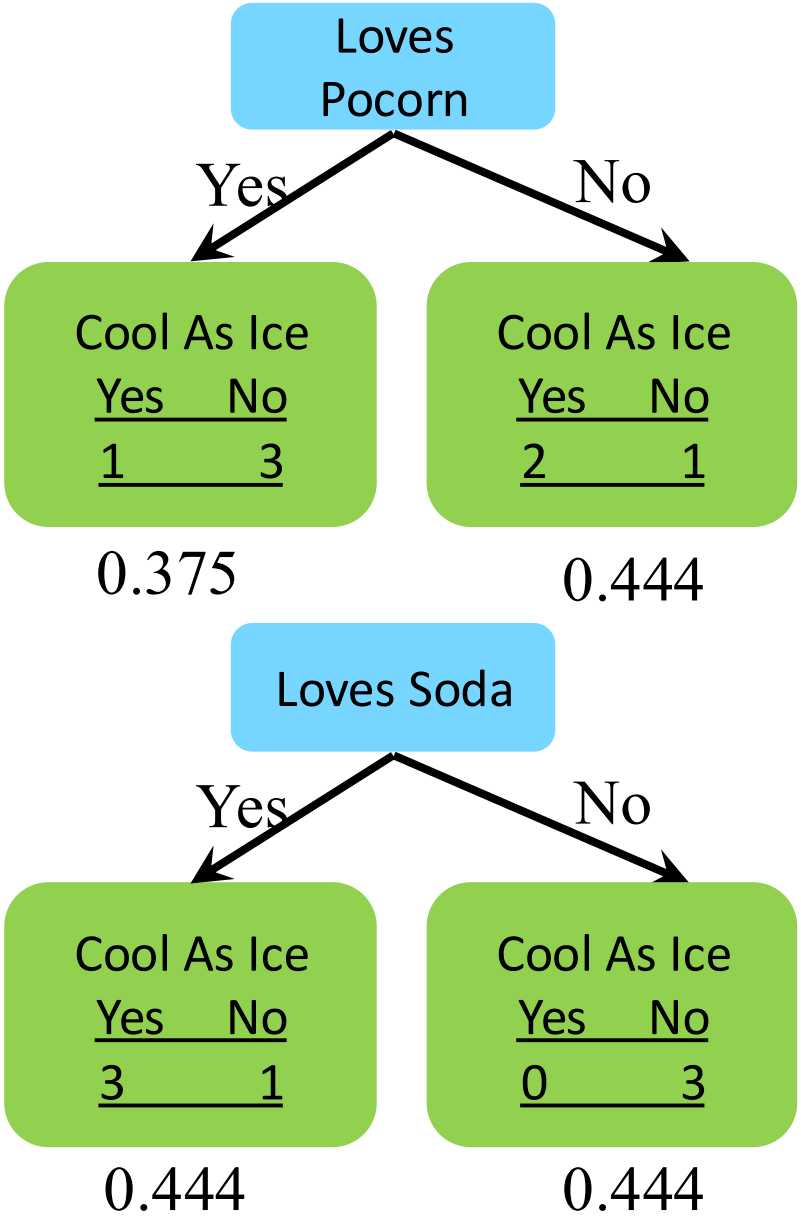
Gini impurity for leaf **Loves popcorn No**

= _____

Gini impurity for leaf **Loves popcorn Yes**

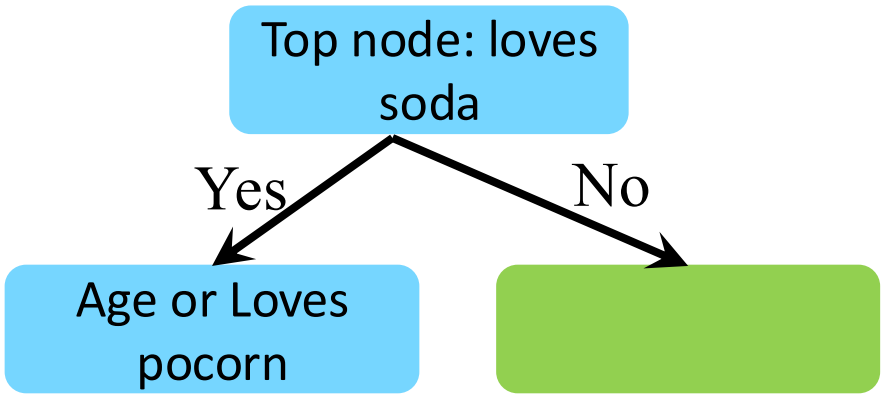
= weighted average of Gini Impurities for the Leaves

=



Example: Predicting "Loves Cool Ice"

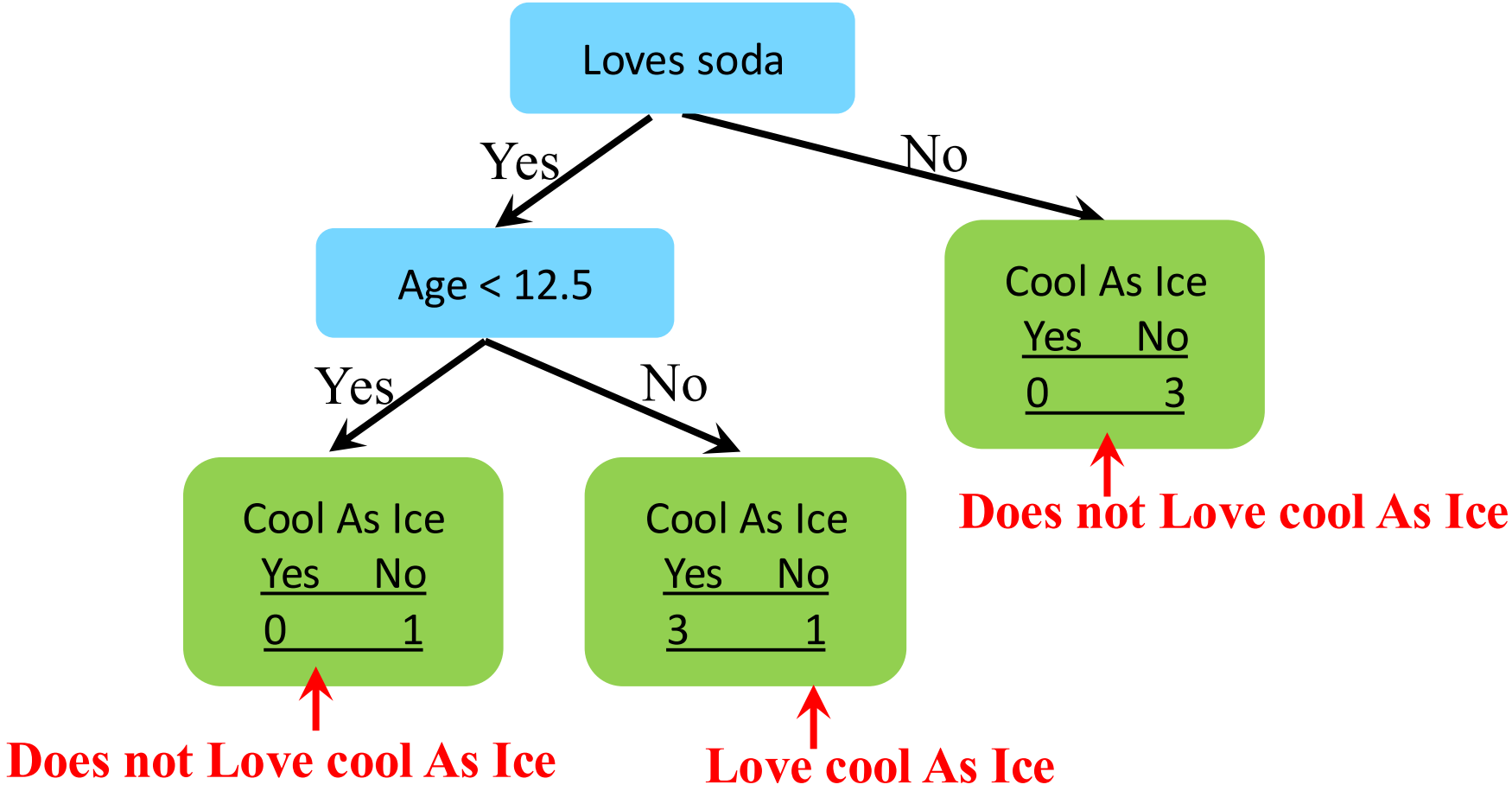
Gini impurity for **Loves popcorn** = 0.405
Gini impurity for **Loves soda** = 0.214
Gini impurity for **Age**= 0.343



- Focus on the current node (e.g., **Love Soda = Yes**)
- Consider remaining features (e.g., **Age**, **Loves Popcorn**)
- For each feature: Compute the **Gini impurity** of the resulting groups
- **Select the feature with the lowest Gini impurity** (most pure split)

  Age is selected (Age = 0 , Popcorn = 0.25)

Example: Predicting "Loves Cool Ice"

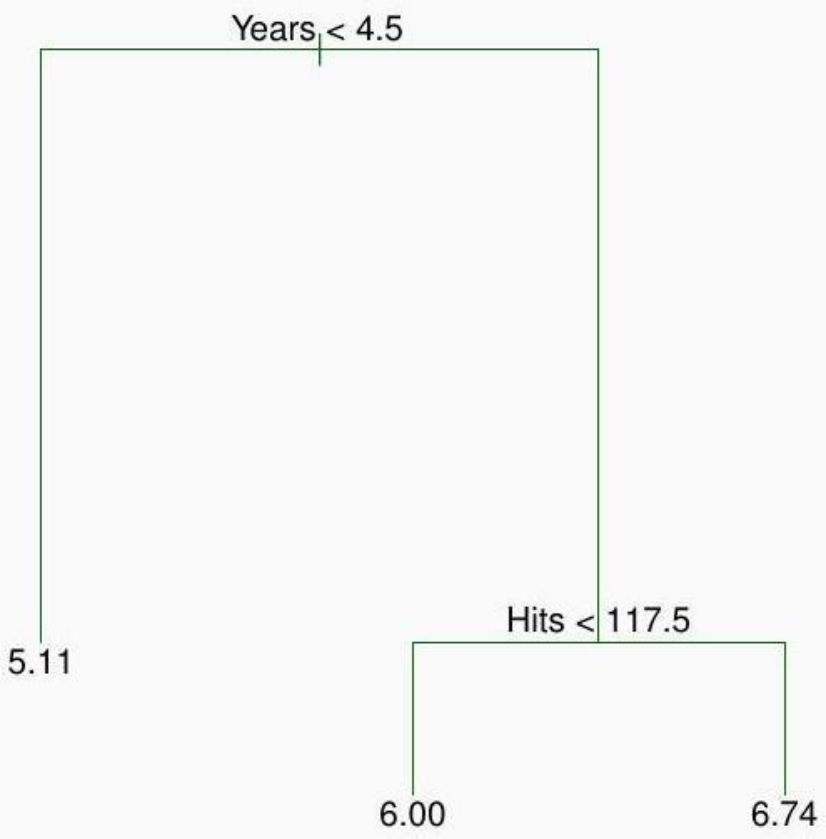


Predict:

Loves Popcorn	Loves Soda	Age	Loves Cool As Ice
Yes	Yes	15	????

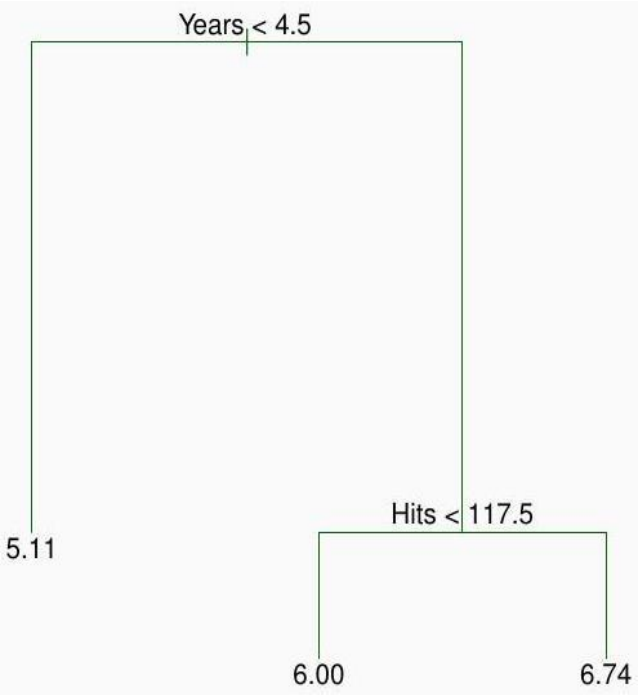
Regression Trees

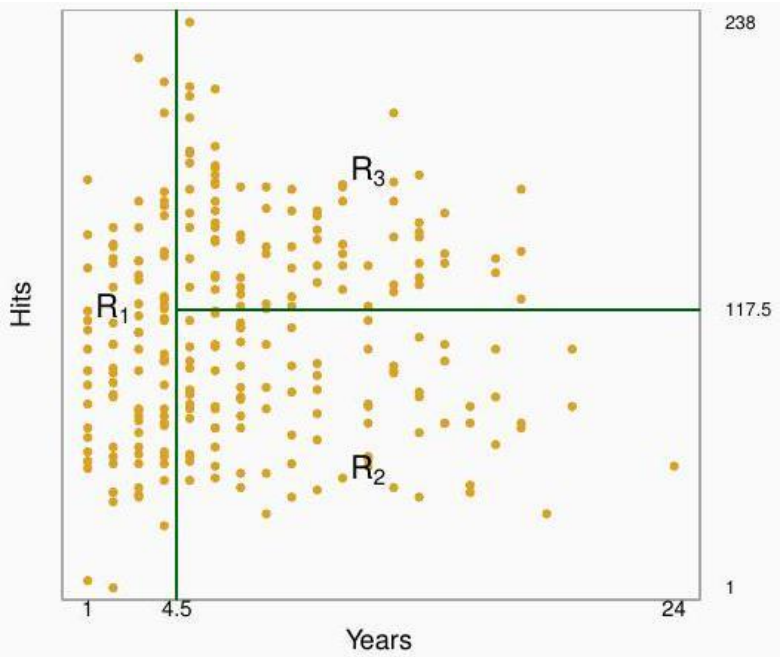
- Example: Hitters dataset. We want to predict a baseball player's **Salary** based on Years and Hits. Salary here is log-transformed so it is more bell-shaped.
- The tree segments (or segments) players into 3 regions of the predictor space:



Regression Trees

- Regions R_1, R_2, R_3 are known as terminal nodes or leaves of the tree.
- Points along the tree where the predictor space is split are referred to as internal nodes.
- Segments of the trees that connect the nodes are known as branches.





Prediction via Segmentation of the Predictor Space

Regression tree building process:

1. We divide the predictor space (the set of possible values for $X_1, X_2, X_3, \ldots, X_p$, into J distinct and non-overlapping regions: $R_1, R_2, \ldots, R_J$.
2. For every observation that falls into the region R_j, we make the same prediction, which is simply the **mean of the response values** for the training observations in R_j.

How do we construct the regions $R_1, \ldots, R_J$?

- In theory, regions could have any shape.
- However, we choose to divide the predictor space into boxes for simplicity and easy of interpretation.
- Then the goal is to find boxes $R_1, \ldots, R_J$ that minimize the RSS, given by:

$$\sum_{j=1}^{J} \sum_{i \in R_j} \left(y_i - \hat{y}_{R_j}\right)^2,$$

where $\hat{y}_{R_j}$ is the mean response for the training observations within the j th box.

How do we construct the regions $R_1, \ldots, R_J$?

- Computationally infeasible to consider every partition of the feature space into J boxes.

Recursive binary splitting

- Top-down recursive greedy approach.
- Top-down recursive: start with all observations and split into two branches at each level of the tree.
- Greedy: best split is made at each step without looking ahead.

Recursive binary splitting

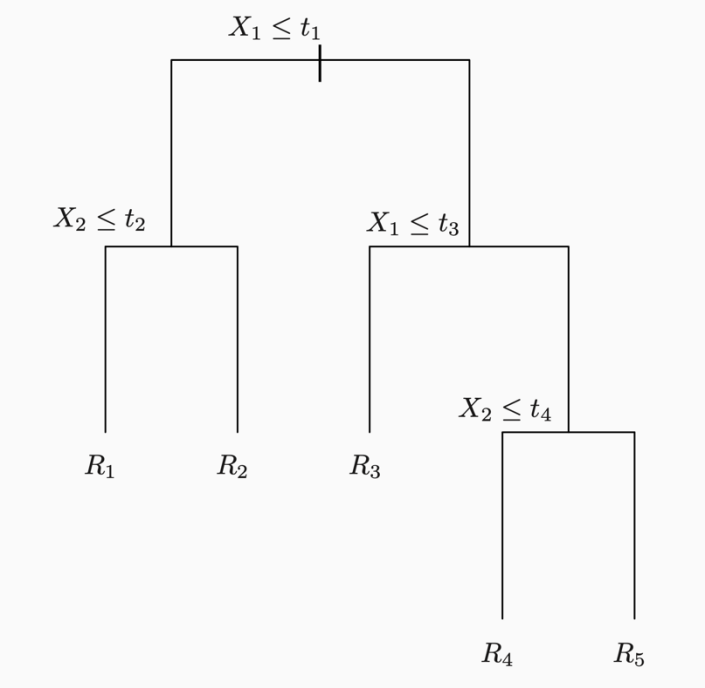
Choose a predictor X_j and a cutpoint s that minimizes the training RSS for the resulting tree:

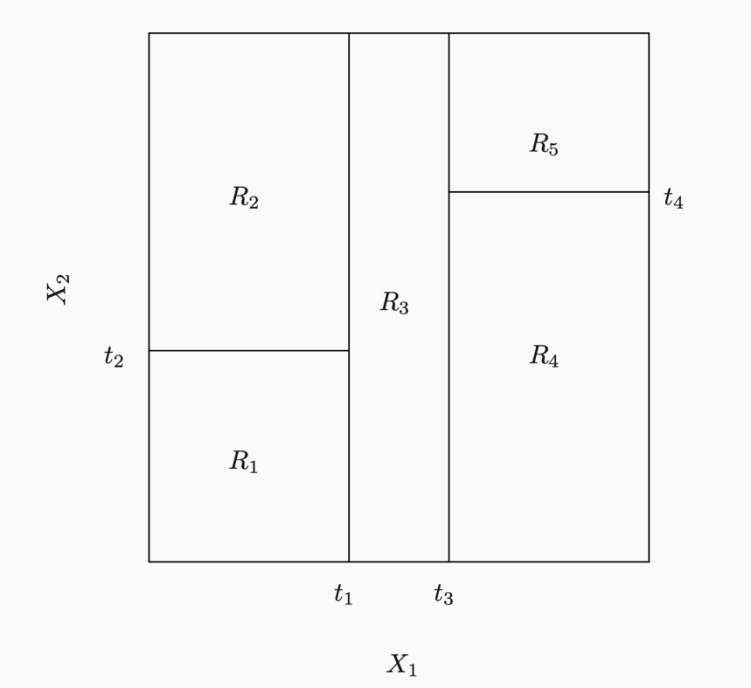
$$R_1(j,s) = \{X \mid X_j < s\}, R_2(j,s) = \{X \mid X_j \geq s\}$$

$$RSS = \sum_{x_i \in R_1(j,s)} \left(y_i - \hat{y}_{R_1}\right)^2 + \sum_{x_i \in R_2(j,s)} \left(y_i - \hat{y}_{R_2}\right)^2$$

Recursive binary splitting

- Repeat the process: look for the best predictor and best cutpoint in order to split the data further so as to minimize the RSS within each resulting region.
- Now instead of splitting entire predictor space, we split one of the previously identified regions.
- This process continues until a stopping criteria is reached (i.e. no regions contains more than 5 observations).





Tree Pruning

- The process we described is a good starting point
- However, it can **overfit** the training data
- This may lead to poor performance on test data

- The tree can become too complex
- Too many splits → too many regions $R_1, R_2, \ldots, R_J$
- The model captures noise instead of true patterns

- Our strategy: **grow a very large tree T_0 and then prune it back in order to obtain a smaller tree (subtree)**.

Cost complexity pruning

- Idea: we apply a **penalty** to the tree.
- For every value of α, there is a subtree $T \subset T_0$ such that:

$$\sum_{m=1}^{|T|} \sum_{i:x_i \in R_m} \left(y_i - \hat{y}_{R_m}\right)^2 + \alpha|T|$$

is as small as possible.

- $|\boldsymbol{T}|$ denotes the number of leaves in the tree.
- $\boldsymbol{\alpha}$ is a tuning parameter: controls the trade-off between the subtree's complexity and its fit to the training data.
- When $\alpha = 0$, we just have the original tree T_0.
- When α increases, what happens to our subtree?
- Reminiscent of lasso.

Algorithm for building a regression tree

1. Use recursive binary splitting to a grow a large tree on the training data, stopping according to a criteria.

2. Apply cost complexity pruning to the large tree in order to obtain a sequence of best subtrees, as a function of α.

3. Use K-fold cross-validation to choose α. For each $k = 1, \ldots, K$:
   a) Repeat Steps 1 and 2 on all but the k th fold of the training data.
   b) Evaluate the mean squared prediction error on the data in the **left-out $\boldsymbol{k}$ th fold**. Average the results for each value of α and pick α to minimize the k-fold CV error.

4. Return the subtree from Step 2 that corresponds to the chosen value of α.

See R script: decisiontreesbasics.R